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B032426 (B255) - ARCHITECTURE, MODEL AND ANALYSIS OF CYBER PHYSICAL SYSTEMS

Bayesian Networks

Lecture 4

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Recap from last lecture

You learned how to:

- Describe the architecture, functionalities, and main failure modes of a modern computer-based railway control system
- Combine multiple modeling formalism to model different aspects/subsystems
- Leverage on the BN formalism for modeling main failure-modes

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Assignment: ERTMS simplified

- Using the parameters from the paper when necessary/appropriate, build a BN model of a simplified ERTMS track with only 1 RBC and 3 trains (read the paper for further details and parameters)
- Implement and evaluate your model in Netica.
- Prepare a few slides Powerpoint presentation about the model and results to be shown and discussed during next lecture.
- Working in small groups is allowed.
- Start working in classroom, then continue at home.
- Assignment is for learning purposes only, and therefore not graded.

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Learning objectives of this lecture

At the end of this lecture you will be able to:

- Use BN model for NMR computer system safety evaluation
- Transfer similar approaches to model other critical CPS components and applications

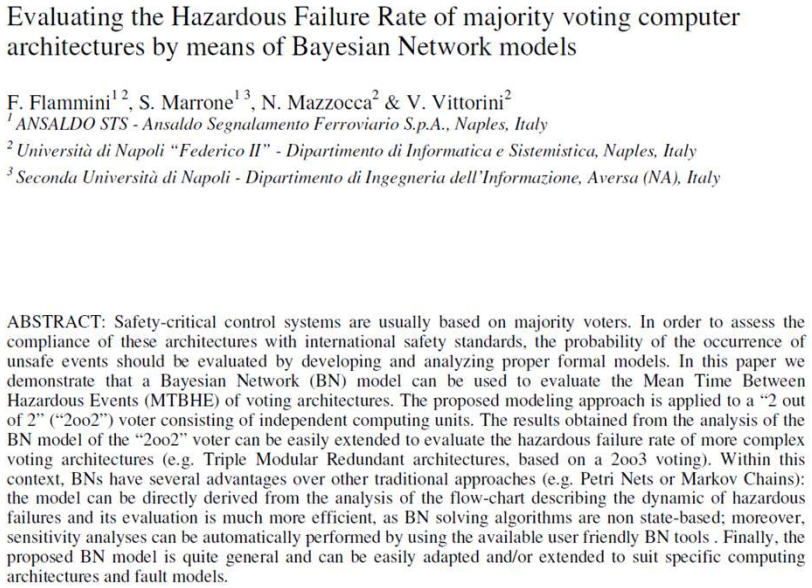
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Learning materials for this lecture

F. Flammini, S. Marrone, N. Mazzocca, and V. Vittorini, "Evaluating the Hazardous Failure Rate of majority voting computer architectures by means of Bayesian Network models," in *Proc. European Safety and Reliability Conference, ESREL*, 2007

Free full-text download from:

https://www.researchgate.net/publication/203738711_Evaluating_the_Hazardous_Failure_Rate_of_majority_voting_computer_architectures_by_means_of_Bayesian_Network_models



Evaluating the Hazardous Failure Rate of majority voting computer architectures by means of Bayesian Network models

F. Flammini[1,2], S. Marrone[1,3], N. Mazzocca[2] & V. Vittorini[2]
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[2] *Università di Napoli "Federico II" - Dipartimento di Informatica e Sistemistica, Naples, Italy*
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ABSTRACT: Safety-critical control systems are usually based on majority voters. In order to assess the compliance of these architectures with international safety standards, the probability of the occurrence of unsafe events should be evaluated by developing and analyzing proper formal models. In this paper we demonstrate that a Bayesian Network (BN) model can be used to evaluate the Mean Time Between Hazardous Events (MTBHE) of voting architectures. The proposed modeling approach is applied to a "2 out of 2" ("2oo2") voter consisting of independent computing units. The results obtained from the analysis of the BN model of the "2oo2" voter can be easily extended to evaluate the hazardous failure rate of more complex voting architectures (e.g. Triple Modular Redundant architectures, based on a 2oo3 voting). Within this context, BNs have several advantages over other traditional approaches (e.g. Petri Nets or Markov Chains): the model can be directly derived from the analysis of the flow-chart describing the dynamic of hazardous failures and its evaluation is much more efficient, as BN solving algorithms are non state-based; moreover, sensitivity analyses can be automatically performed by using the available user friendly BN tools . Finally, the proposed BN model is quite general and can be easily adapted and/or extended to suit specific computing architectures and fault models.

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About «soft errors» (transient faults) in computers

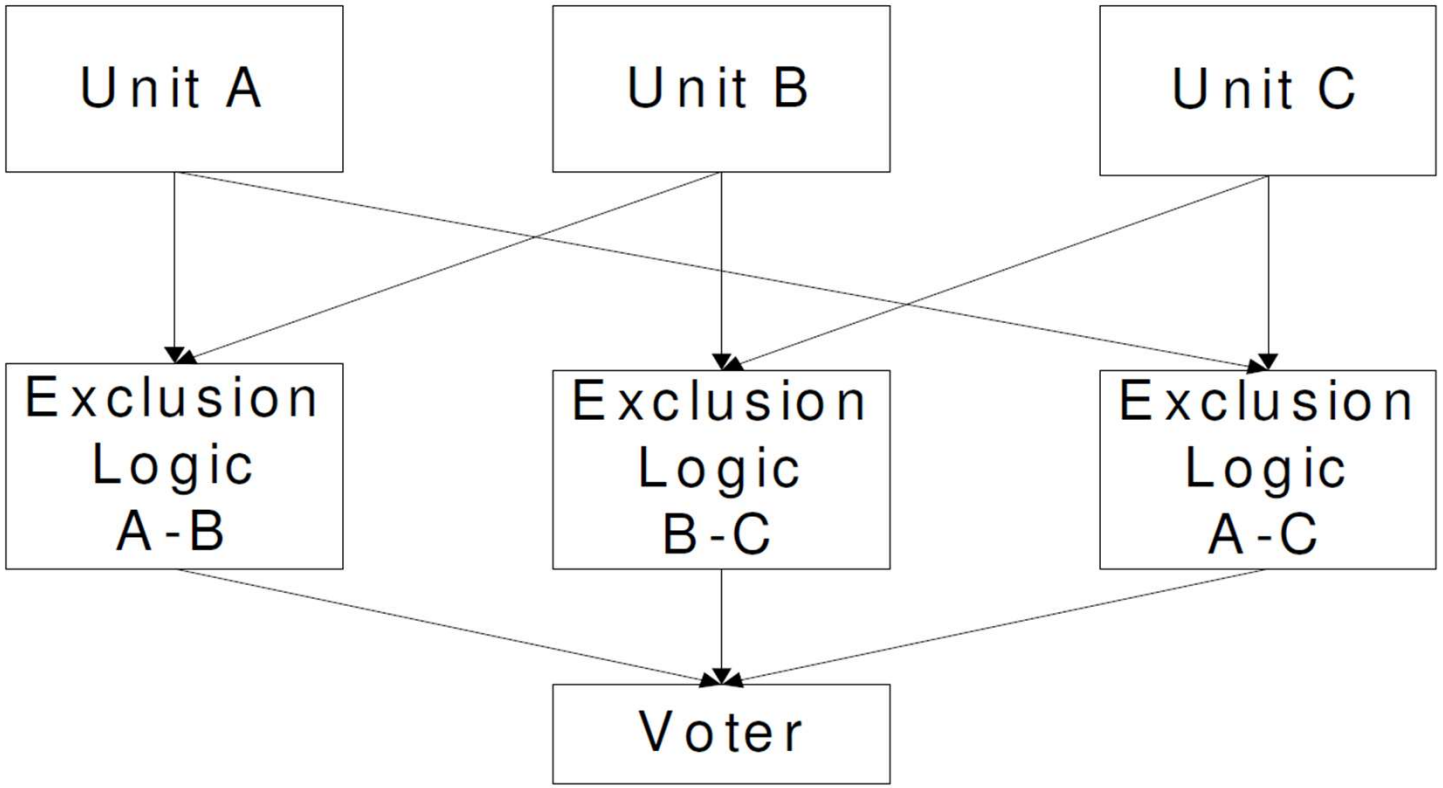
Short video documentary:

"The Universe is Hostile to Computers"

https://www.youtube.com/watch?v=AaZ_RSt0KP8

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Triple Modular Redundant (TMR) computer architecture

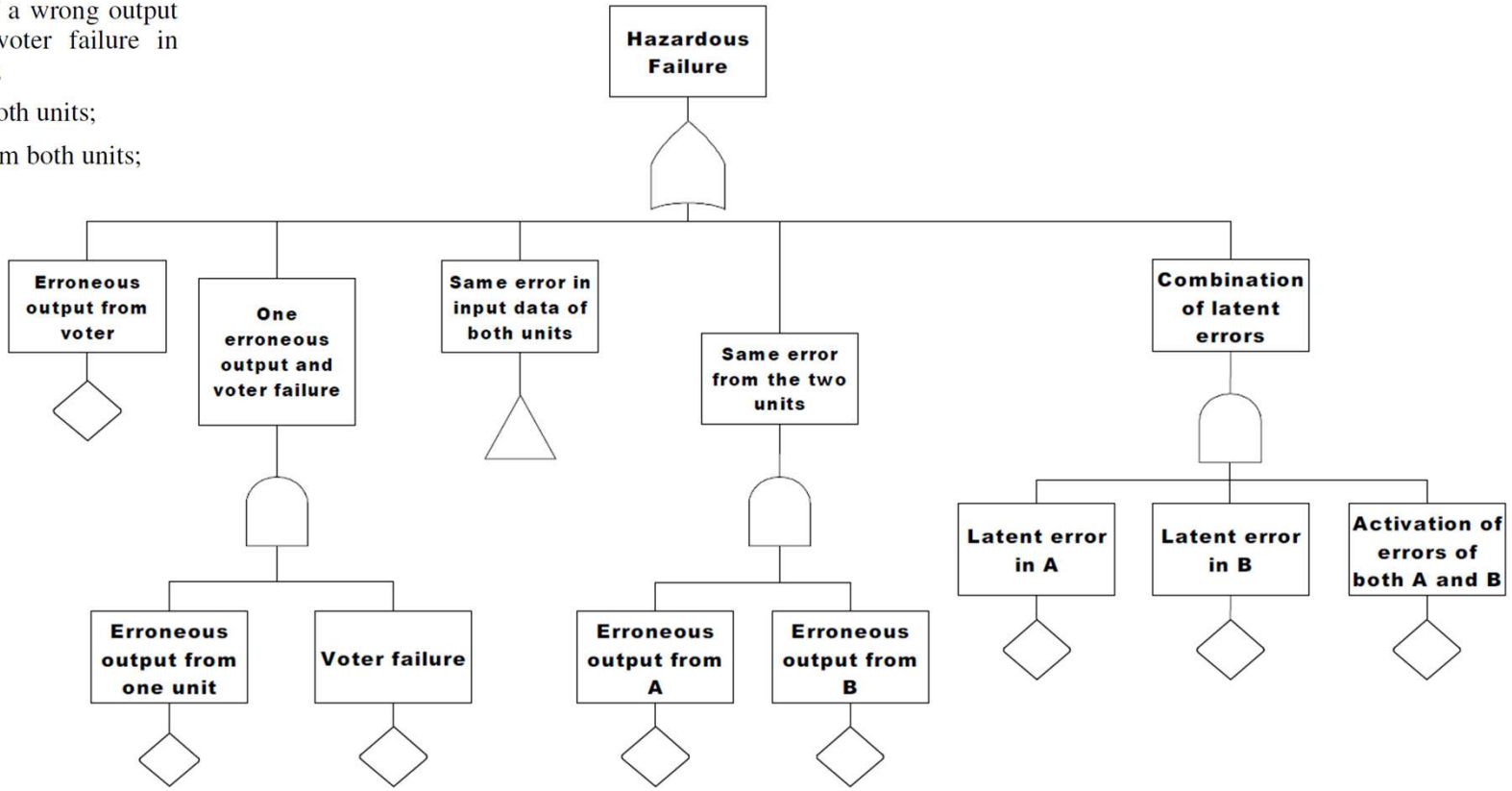


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Qualitative FT model for the 2oo2 hazardous failure

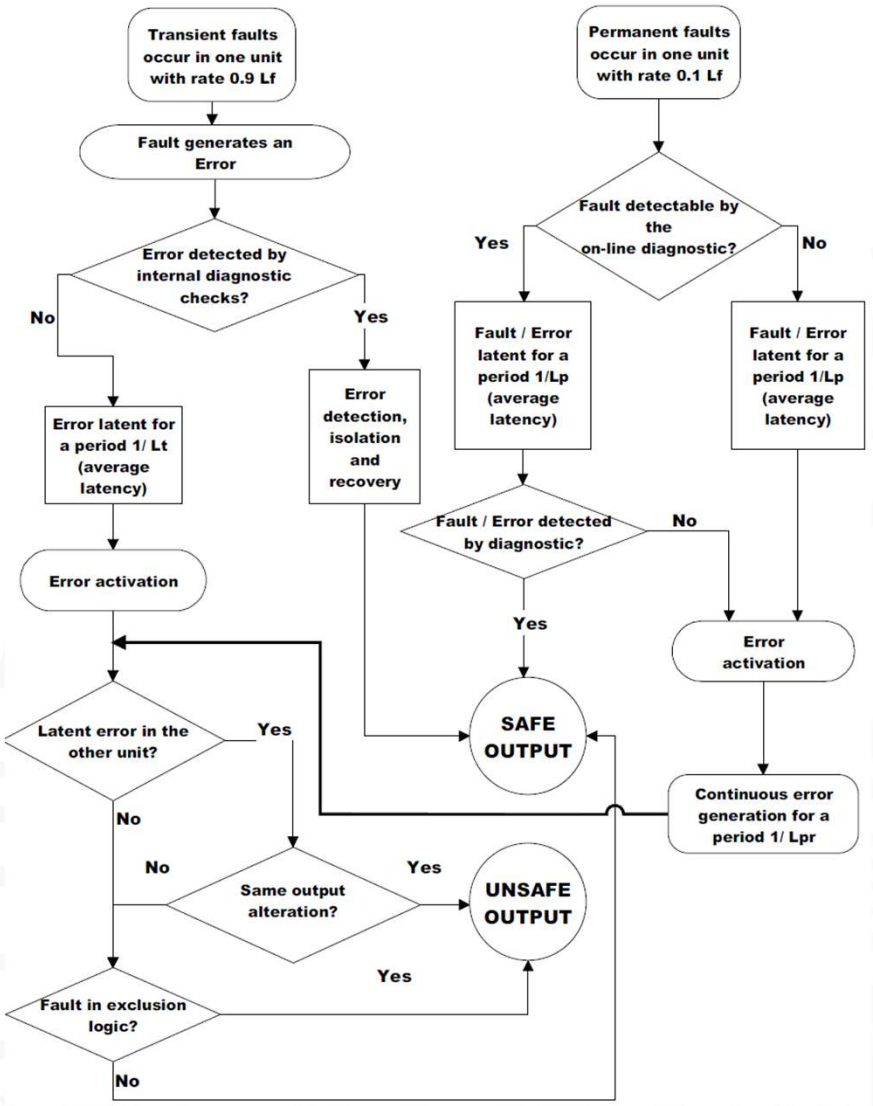
1. fault in the voter which affects the output;
2. simultaneously occurrence of a wrong output of one faulty unit and a voter failure in detecting unmatching outputs;
3. undetected invalid inputs to both units;
4. undetected invalid outputs from both units;

(1) and (3) are negligible



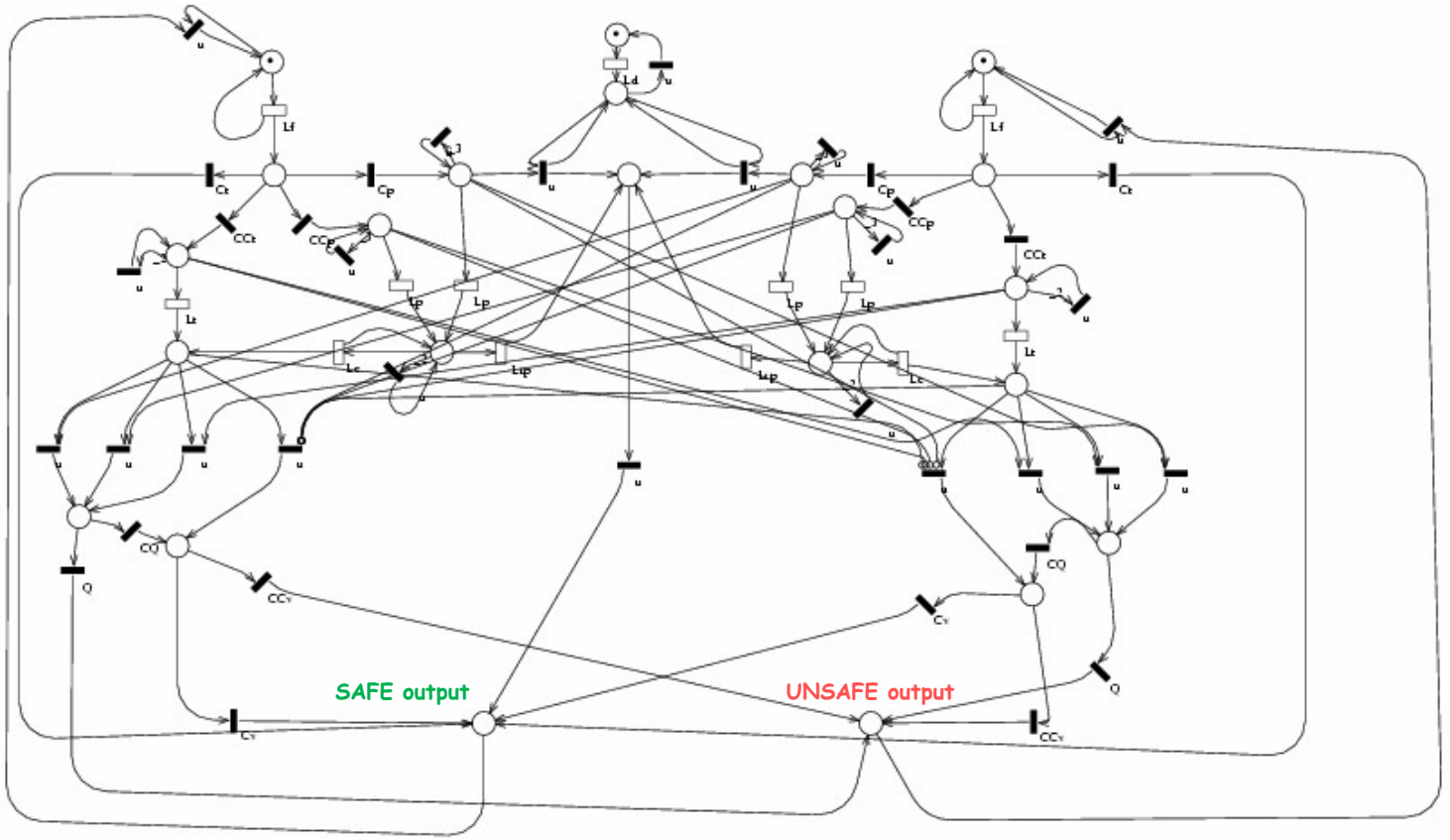
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Hazardous failure flow-chart



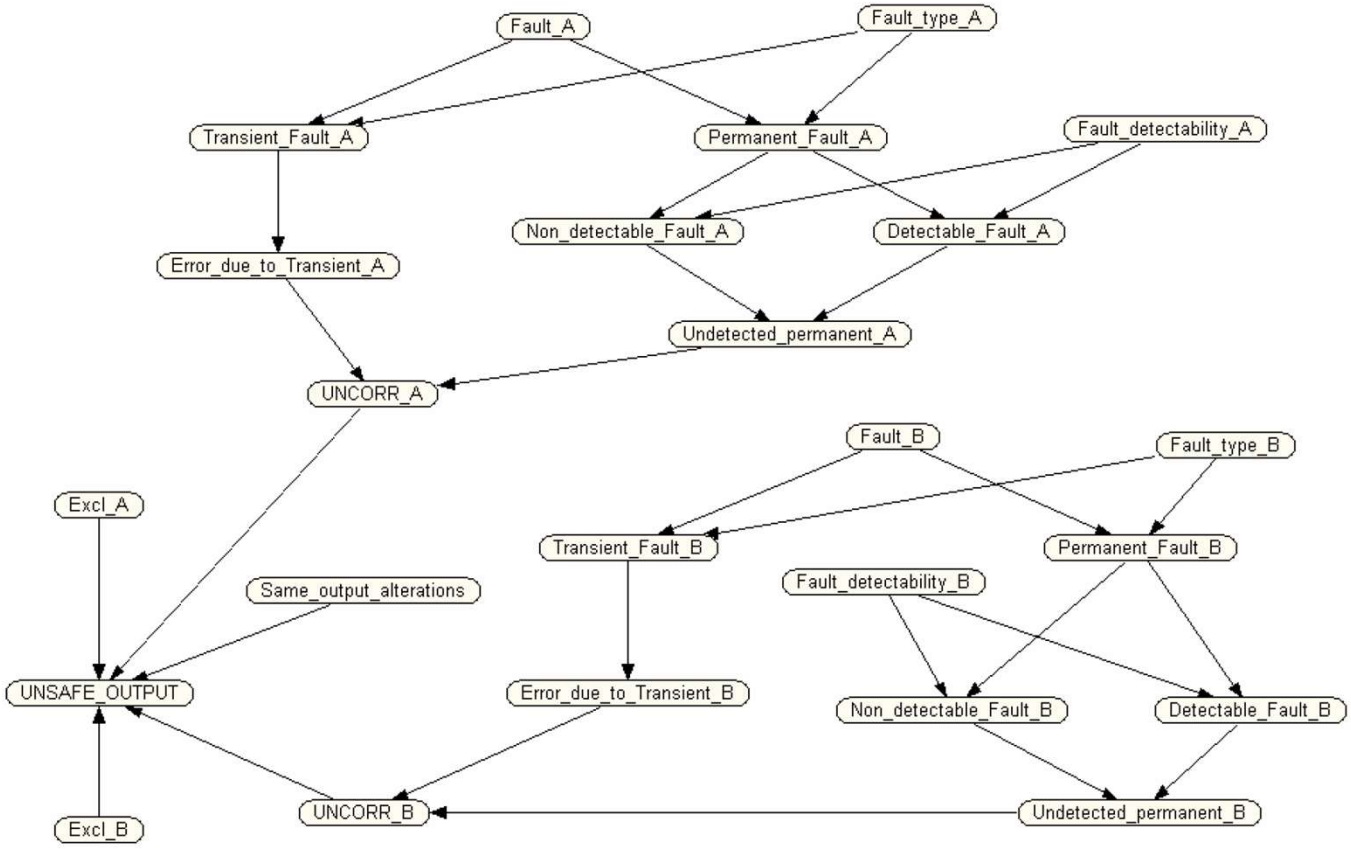
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How a Stochastic Petri Net model would look like



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BN model for hazardous failure in 2oo2



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CPT for UNSAFE OUTPUT

True	False	Same Output Alt.	UNCORR_A	UNCORR_B	Excl_A	Excl_B
1	0	True	True	True	True	True
					True	False
					False	True
					False	False
				False	True	True
					True	False
					False	True
					False	False
0	1					
0	1					
1	0		False	True	True	True
					True	False
					False	True
					False	False
0	1			False	True	True
					True	False
					False	True
					False	False
1	0	False	True	True	True	True
					True	False
					False	True
					False	False
1	0			False	True	True
					True	False
					False	True
					False	False
0	1					
0	1					
1	0		False	True	True	True
					True	False
					False	True
					False	False
0	1			False	True	True
					True	False
					False	True
					False	False

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BN parameters and results

STOCHASTIC VAR.	MEANING	VALUE	PROBABILITY
Detectable_Fault_A	A detectable fault occurring in Unit A	True	1.4999e-006
Detectable_Fault_B	A detectable fault occurring in Unit B	True	1.4999e-006
Error_due_to_Transient_A	An error due to a transient fault occurring in Unit A	True	1.4995e-006
Error_due_to_Transient_B	An error due to a transient fault occurring in Unit B	True	1.4995e-006
Excl_A	A failure occurring in the Exclusion Logic of Unit A	True	1e-010
Excl_B	A failure occurring in the Exclusion Logic of Unit B	True	1e-010
Fault_A	A fault occurring in Unit A	True	1.6666e-005
Fault_B	A fault occurring in Unit B	True	1.6666e-005
Fault_detectability_A	The possibility of detecting a fault occurring in Unit A	Detectable	0.9
Fault_detectability_B	The possibility of detecting a fault occurring in Unit B	Detectable	0.9
Fault_type_A	The fault type of Unit A (Transient or Permanent)	Transient	0.9
Fault_type_B	The fault type of Unit B (Transient or Permanent)	Transient	0.9
Non_detectable_Fault_A	A non detectable fault occurring in Unit A	True	1.6666e-007
Non_detectable_Fault_B	A non detectable fault occurring in Unit B	True	1.6666e-007
Permanent_Fault_A	A permanent fault occurring in Unit A	True	1.6666e-006
Permanent_Fault_B	A permanent fault occurring in Unit B	True	1.6666e-006
Same_output_alterations	The two units produce the same modification of their output	True	0.1
Transient_Fault_A	A transient fault occurring in Unit A	True	1.4999e-005
Transient_Fault_B	A transient fault occurring in Unit B	True	1.4999e-005
UNCORR_A	An uncorrect output occurring in Unit A	True	2.1912e-006
UNCORR_B	An uncorrect output occurring in Unit B	True	2.1912e-006
Undetected_permanent_A	An undetected permanent fault occurring in Unit A	True	6.9164e-007
Undetected_permanent_B	An undetected permanent fault occurring in Unit B	True	6.9164e-007
UNSAFE_OUTPUT	**An unsafe output occurring in the system**	**True**	**4.8056e-013**

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Computing MTBHE of the TMR

The "2oo3" failure model can be derived from the "2oo2" model:

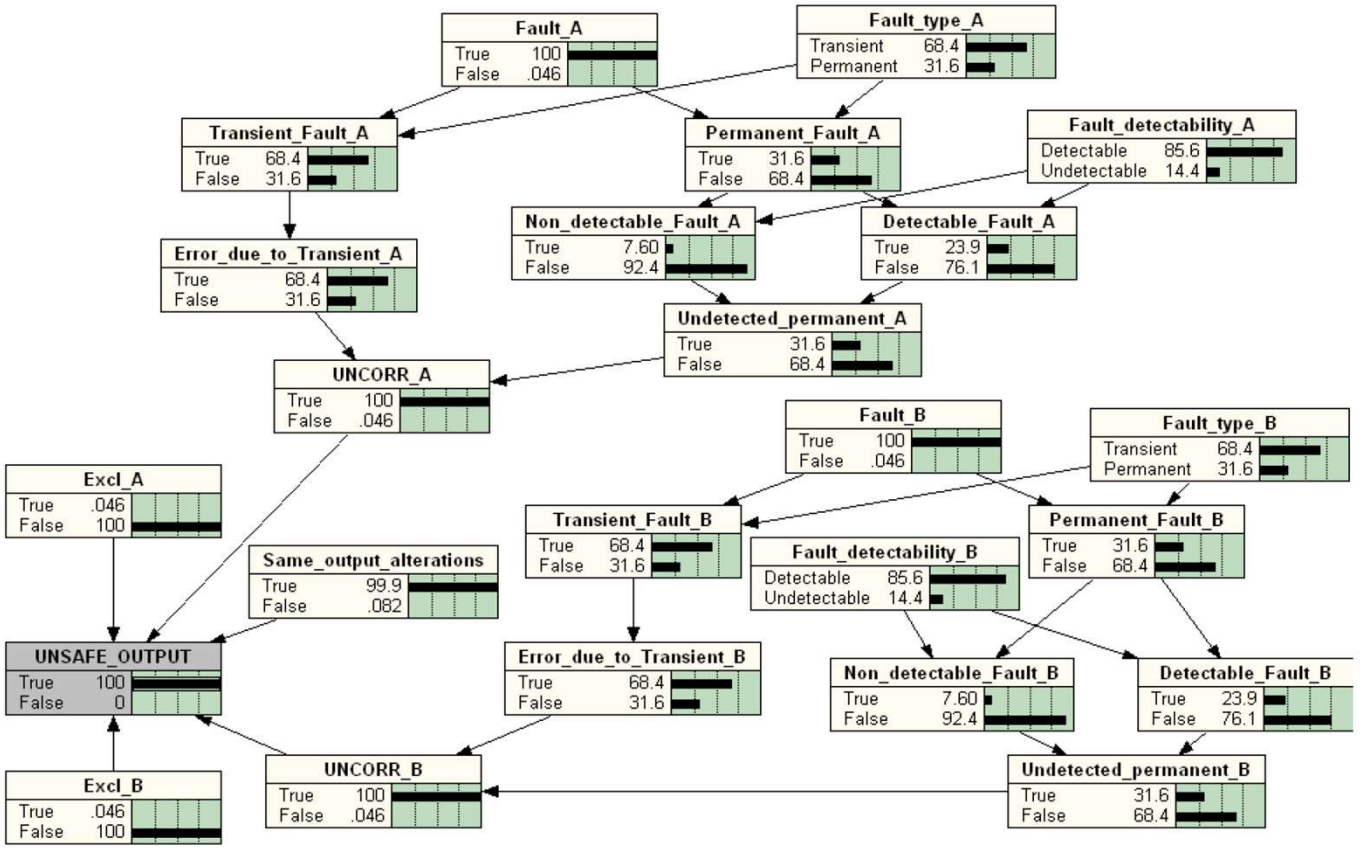
$$MTBHE_{2oo3} = 1/3 \cdot MTBHE_{2oo2}$$

In fact, the behavior of a "2oo3" system can be described by three "2oo2" systems: units A AND B, units B AND C, units A AND C.

$$MTBHE_{2oo3} = 1 / (3 \cdot HR_{2oo2}) = 6.9362 \cdot 10^{11} \text{ h}$$

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A posteriori probabilities



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Generalizing the approach to diverse redundant detectors

- The output of modern, possibly «smart», sensing devices can be combined to improve system safety in critical CPS applications such as industrial control, infrastructure monitoring, and perception systems in autonomous robots such as self-driving vehicles.
- Technological diversity assumption is essential but correlations due to common mode of failures (e.g., environmental disturbances affecting more sensors) can be modelled to some extent.
- Due to their ease of use and intrinsic explainability, BNs can be very useful in those applications.

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Further reading

- F. Flammini. Model-based analysis of ‘k out of m’ correlation techniques for diverse redundant detectors. International Journal of Performability Engineering, 9(5):551, 2013. https://www.ijpe-online.com/EN/10.23940/ijpe.13.5.p551.mag
- F. Flammini, S. Marrone, R. Nardone, M. Caporuscio, and M. D’Angelo. Safety integrity through self-adaptation for multi-sensor event detection: Methodology and case-study. Future Generation Computer Systems, 112:965–981, 2020. https://www.researchgate.net/publication/342486439_Safety_integrity_through_self-adaptation_for_multi-sensor_event_detection_Methodology_and_case-study
- M. J. Pappaterra and F. Flammini, "A Review of Intelligent Cybersecurity with Bayesian Networks," *2019 IEEE International Conference on Systems, Man and Cybernetics (SMC)*, Bari, Italy, 2019, pp. 445-452, doi: 10.1109/SMC.2019.8913864

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Exam assignment (1)

AMA-CPS Bayesian Network exam assignment: mini-project

Individual work on the topics addressed in the Bayesian Network (BN) part of the course. The project will contribute to the final score through a weighted average. Please follow the instructions below.

1. Choose a CPS of your choice and a suitable sub-system to be modelled.
2. Define the dependability attributes of interest and related quantitative requirements.
3. Assume credible threat metrics (e.g., MTBF) for the basic components, and other possible external events.
4. If you start from Fault Tree model, justify the need for adding any common mode of failure, and/or any noisy-gates. Define the CPTs accordingly.
5. Build the BN model using a suitable tool such as Netica. It is recommended that the model should not exceed the size of 15 nodes.
6. Analyze the BN model using the techniques shown during the lectures.
7. Report and discuss the results.

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Exam assignment (2)

You are expected to write a brief report on those points (suggested length between 5 and 10 pages; no special requirements about the format or template), and a short presentation between 4 and 6 slides. Please submit the source files of your BN model as well as the report and the presentation (in pdf format).

Submission and deadlines

Information on how and when to submit the materials will be provided on the specific section of the course website. In case of any issues during submission, please email the docent.

You must be prepared to present your slides during an oral exam day scheduled after mini-project submission deadline, if required.

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Exam assignment (3)

Evaluation criteria

Your BN mini-project will be evaluated according to the following criteria:

1. Clarity and logic in the overall presentation of the use-case.
2. Novelty and significance of the application in relation to course subject coverage (i.e., CPS).
3. Complexity of the BN, although size not should exceed 15 nodes as stated above.
4. Correctness and accuracy in the structure and parameters of the BN model.
5. Relevance and completeness of the analysis performed on the BN model.
6. Quality and depth in the discussion of the results.

Each criterion weighs 5 points. A minimum of 2 points for the score of each criterion, and a minimum of 18 points as the sum of the scores for all criteria are needed to pass the exam.

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Thank you for your attention

Questions?

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